

CSE141 INTRODUCTION TO PROGRAMMING

Second Term Examination

Max Marks: 35

Time Allowed: 1½ hours

Answer the questions in the spaces provided on the question sheets.
Please give clear and rigorous answers. Be to the point. Show your work.

Name: _____

ERP: _____

Page:	1	2	3	4	5	6	Total
Marks:	5	10	4	6	6	4	35
Score:							

Question 1: 5 marks

(a) [3 marks] Write statements to do the following:

i. Create an array to hold 10 double values.

.....

ii. Assign the value 5.5 to the last element in an array of doubles.

.....

iii. Use an array initializer to create another array with the initial values 3.5, 5.5, 4.52, and 5.6.

.....

(b) [2 marks] Give the sequence of integers printed by a call to print(7):

```
public static void print(int n) {  
    if (n <= 0) return;  
    StdOut.print(n + " ");  
    print(n-2);  
    StdOut.print(n + " ");  
}
```

.....
.....
.....

Question 2: 10 marks

Each of the following code snippets contains a print statement. However, some of them contain errors. For each of the snippets below, either categorize the error as a compilation error or runtime error, or, if there is no error, provide the output of the program.

A.	<pre>int[] a = new int[5]; System.out.println(a[5]);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____
B.	<pre>int[] a = new int[4]; System.out.println(a.length);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____
C.	<pre>int[] a = {1, 2, 3}; int[] b = {1, 2, 3}; System.out.println(a == b);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____
D.	<pre>int[] a = {2, 4, 6, 8, 10}; int[] b = a; System.out.println(b[3]);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____
E.	<pre>int[] a; System.out.println(a[3]);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____
F.	<pre>int[] a = {1, 2, 3, 4, 5}; a = new int[4]; for (int i = 0; i < a.length; i++) a[i] = 2 * i; System.out.println(a[3]);</pre>	☐ compilation error ☐ runtime error ☐ no error; the output is: _____

Question 3: 4 marks

- (a) [2 marks] Write a method called `swapPairs` that accepts an array of integers and swaps the elements at adjacent indexes. That is, elements 0 and 1 are swapped, elements 2 and 3 are swapped, and so on. If the array has an odd length, the final element should be left unmodified. For example, the list `{10, 20, 30, 40, 50}` should become `{20, 10, 40, 30, 50}` after a call to your method.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) [2 marks] Assume that a two-dimensional rectangular array of integers called `data` has been declared with four rows and seven columns. Write a loop to initialize the third row of `data` to store the numbers 1 through 7.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 4:6 marks

- (a) [3 marks] Write a method called `append` that accepts two integer arrays as parameters and returns a new array that contains the result of appending the second array's values at the end of the first array. For example, if arrays `list1` and `list2` store `{2, 4, 6}` and `{1, 2, 3, 4, 5}` respectively, the call of `append(list1, list2)` should return a new array containing `{2, 4, 6, 1, 2, 3, 4, 5}`. If the call instead had been `append(list2, list1)`, the method would return an array containing `{1, 2, 3, 4, 5, 2, 4, 6}`.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) [3 marks] Convert the following iterative method into a recursive method:

```
// Prints each character of the string reversed twice.  
// doubleReverse("hello") prints oolllleehh  
public static void doubleReverse(String s) {  
    for (int i = s.length() - 1; i >= 0; i--) {  
        System.out.print(s.charAt(i));  
        System.out.print(s.charAt(i));  
    }  
}
```

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 5:6 marks

(a) [3 marks] Consider the following method:

```
public static void mystery1(int n) {  
    if (n <= 1) {  
        System.out.print(n);  
    } else {  
        mystery1(n / 2);  
        System.out.print(", " + n);  
    }  
}
```

For each of the following calls, indicate the output that is produced by the method:

- i. `mystery1(1);`
- ii. `mystery1(2);`
- iii. `mystery1(10);`

.....
.....
.....
.....
.....
.....

(b) [3 marks] Consider the following method:

```
public static int mystery2(int x, int y) {  
    if (x < y) {  
        return x;  
    } else {  
        return mystery2(x-y, y);  
    }  
}
```

For each of the following calls, indicate the value that is returned:

- i. `mystery2(6, 13)`
- ii. `mystery2(8, 2)`

.....
.....
.....
.....
.....
.....

Write a method called `writeSequence` that accepts an integer n as a parameter and prints to the console a symmetric sequence of n numbers composed of descending integers that ends in 1, followed by a sequence of ascending integers that begins with 1. The following table indicates the output that should be produced for various values of n :

Notice that when n is odd the sequence has a single 1 in the middle, while for even values it has two 1s in the middle.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.